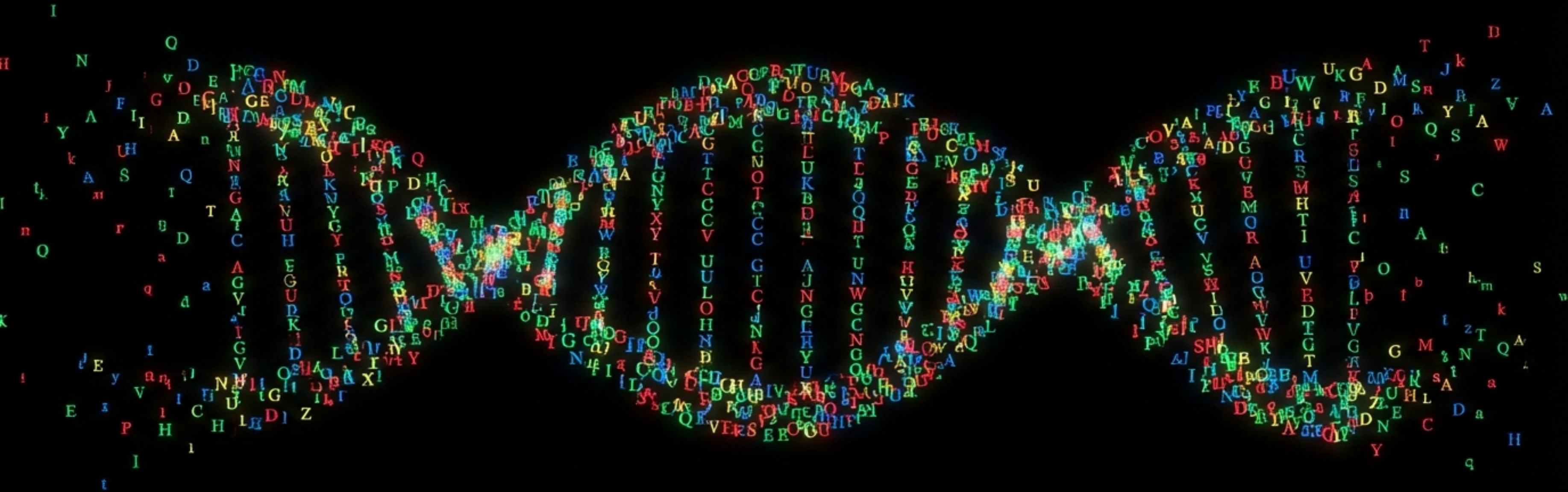


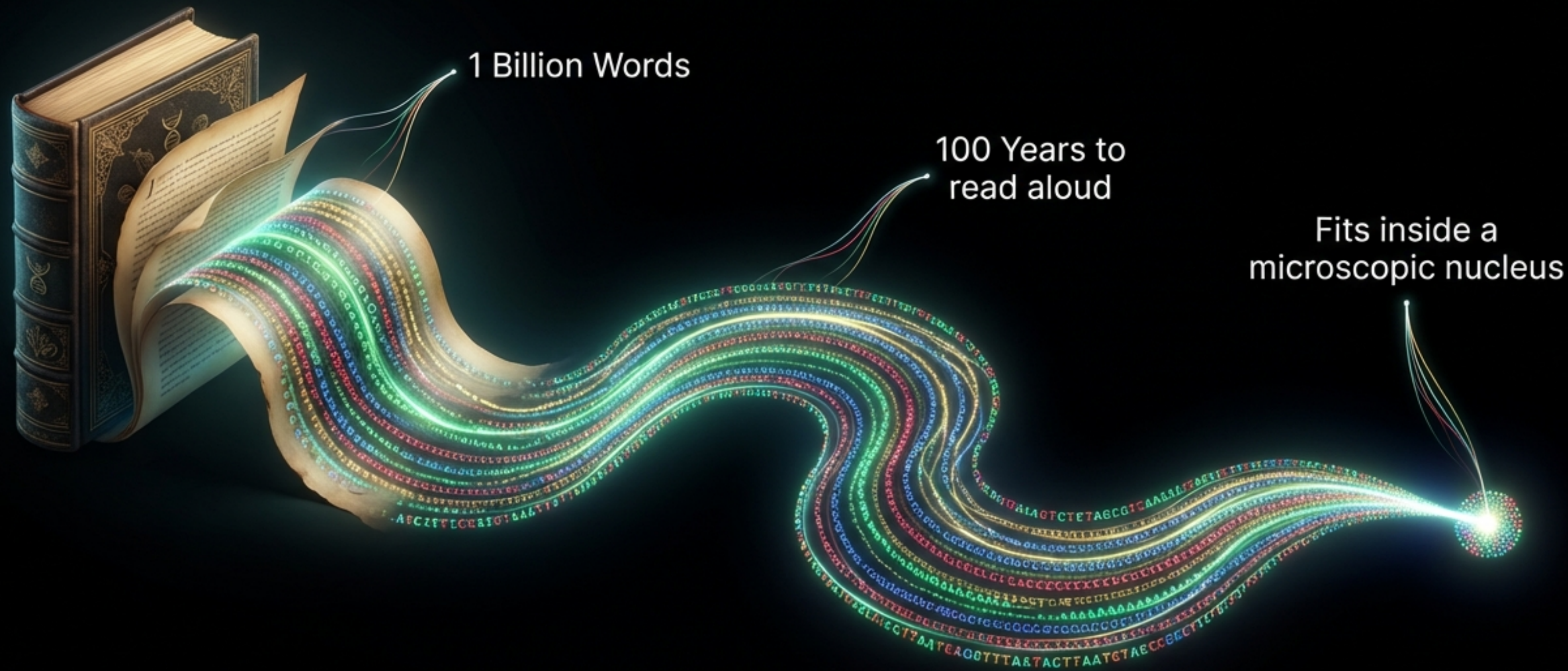
GENOME

The Autobiography of a Species in 23 Chapters



Based on the book by Matt Ridley

A Book the Length of the Danube



A Narrative Glossary

The Biology

The Genome

23 Chromosomes

Genes (30k - 80k)

Exons

Codons

Bases (A, C, G, T)

The Literature

The Book

The Chapters

The Stories

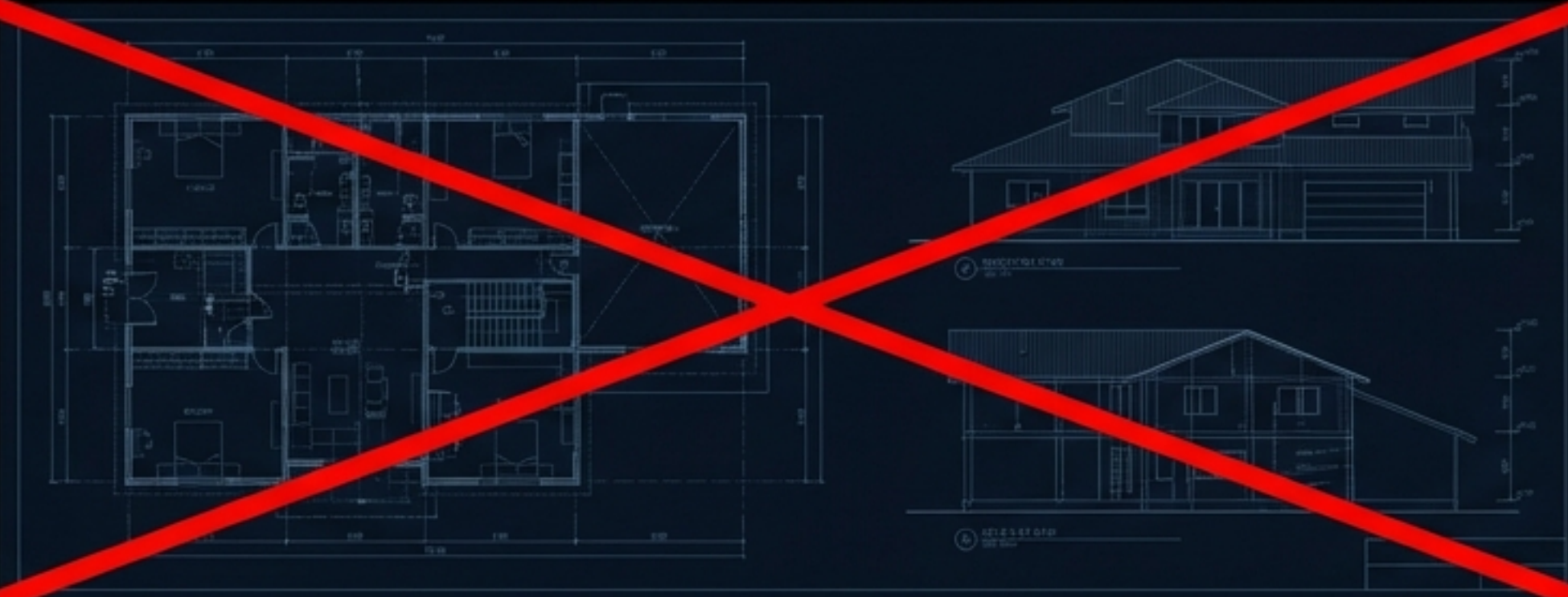
The Paragraphs

The Words

The Letters

The idea of the genome as a book is not a metaphor.
It is literally true. Both are digital information.

It Is Not a Blueprint. It Is a Recipe.



Blueprint: 2D Map. Correspond to parts.

The Genome is a linear digital code. The 'cook' is the cellular machinery that translates the code into a protein.

ADD SUGAR > MIX > BAKE > RESULT: CAKE

Recipe: 1D Instruction. Executed by the cook.

The Photocopy and The Reader

REPLICATION
(The Photocopy)

TRANSLATION
(The Reading)



A pairs with T, G pairs with C.
Information lasts indefinitely.

DNA → RNA → Ribosome → Protein.

"A protein is just a gene's way of making another gene."

Primordial Soup

CHROMOSOME 1

Life

In the beginning was the word.

The Paradox: The Chicken or the Egg?



RNA is the ancestor. It can carry code (like DNA) and act as a catalyst (like Protein).
The "Ur-gene" was a replicator-catalyst.

LUCA: Last Universal Common Ancestor



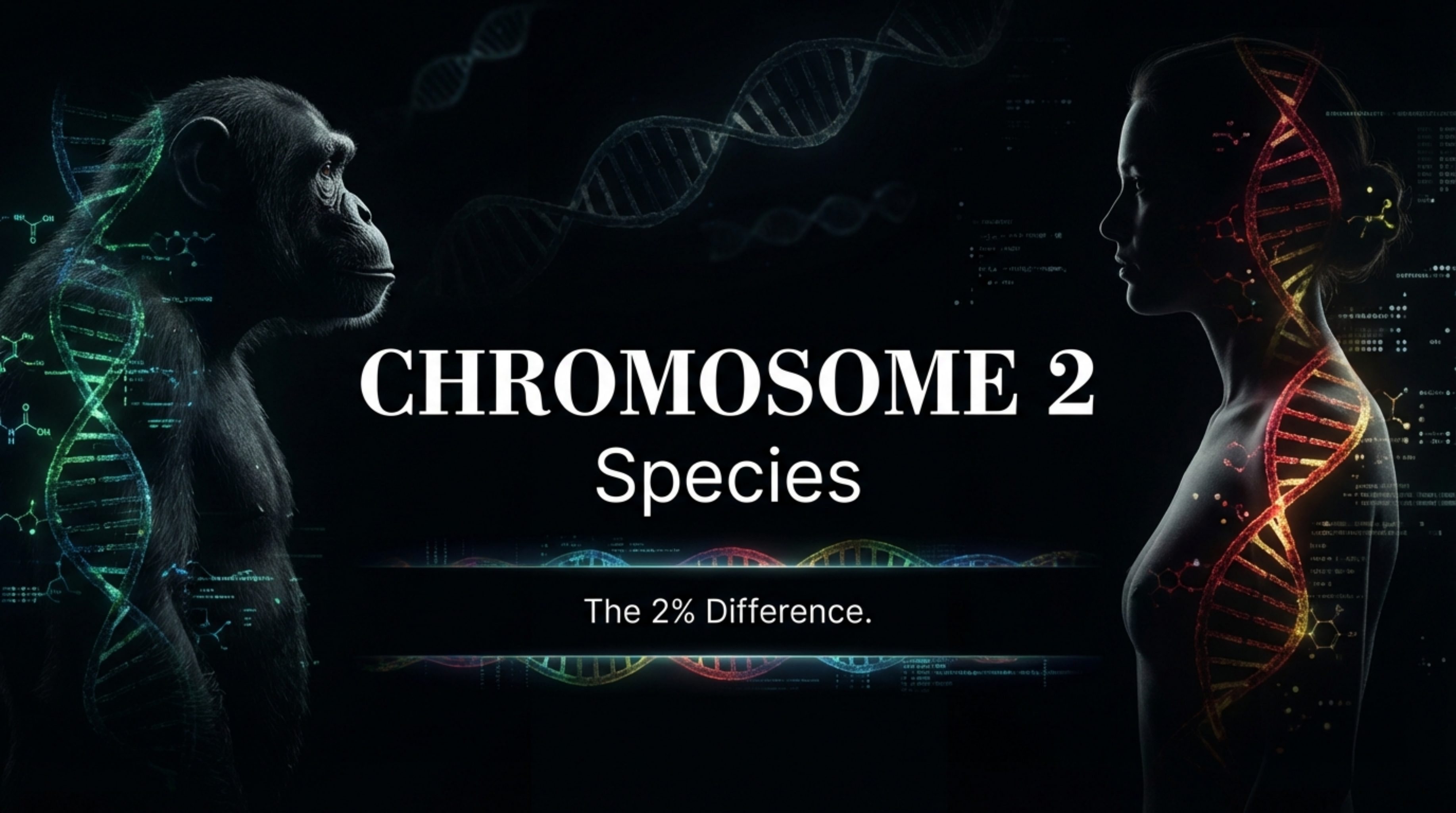
4 Billion Years Ago.



LUCA

The Unity of Life:

The genetic code is universal. "CGA" means arginine in a bat, a beetle, a beech tree, and a bacteria.



CHROMOSOME 2

Species

The 2% Difference.

The Case of the Missing Pair

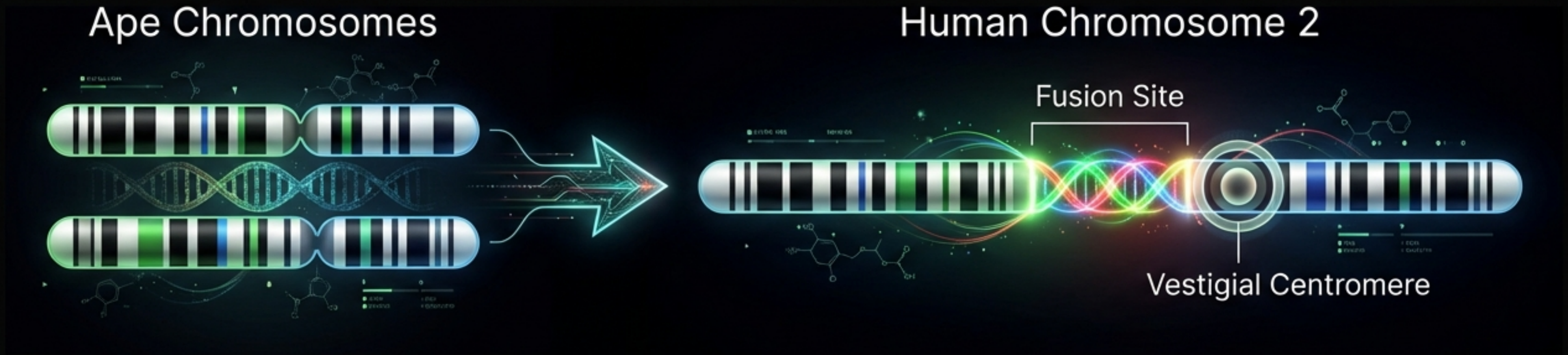
1921: Painter counts 24 pairs.

1955: Tjio counts 23 pairs.

The Reality: Apes have 24 pairs. Humans have 23.

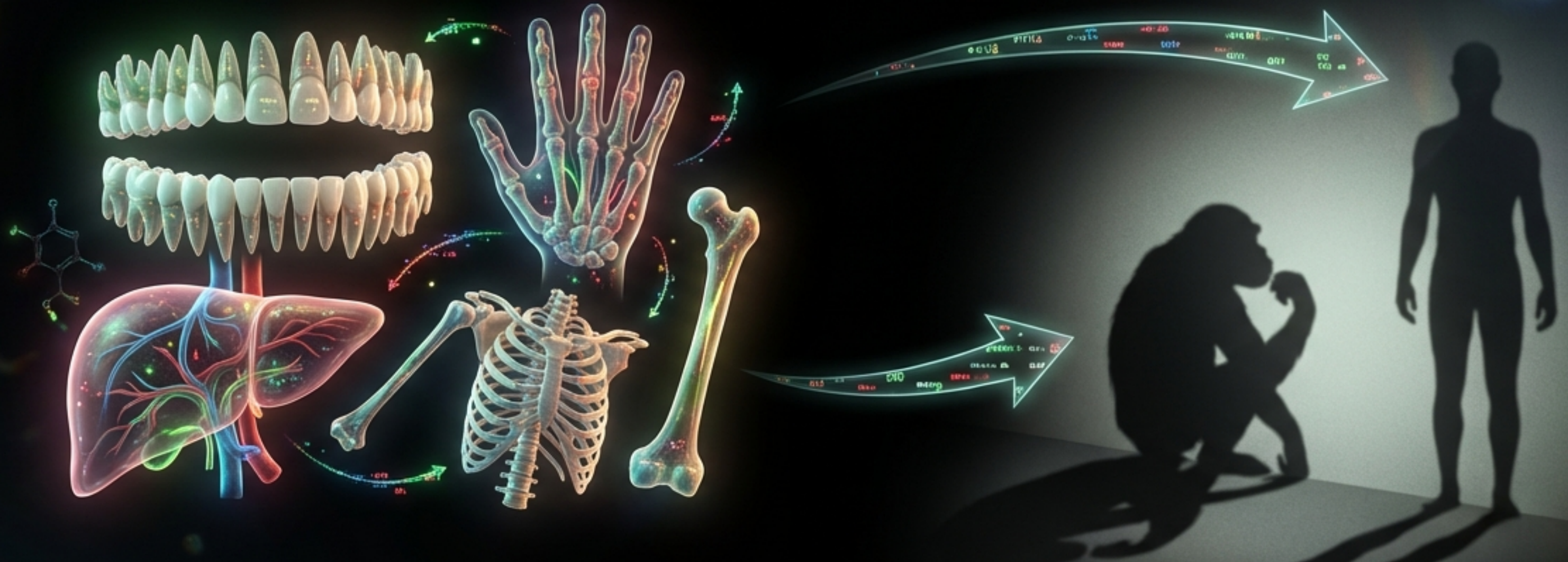
For 30 years, science saw what it expected to see.
We have one fewer pair than our closest cousins.

The Fusion Event



We didn't lose a pair. Two ape chromosomes fused to form Human Chromosome 2. This event may have isolated our ancestors.

We Are 98% Chimpanzee



The Plasticine Analogy: The ingredients are identical. The difference is in the recipe instructions (growth rates, hormones).

In 13 chromosomes, there are zero visible differences.

Hardware vs. Software

Genes (The Negative)

Chimpanzee Nucleus Data



Womb (The Developer)

Human Gestational Environment



If you put a chimp nucleus (negative) into a human womb (developer), you get a chimp. The environment cannot change the species.

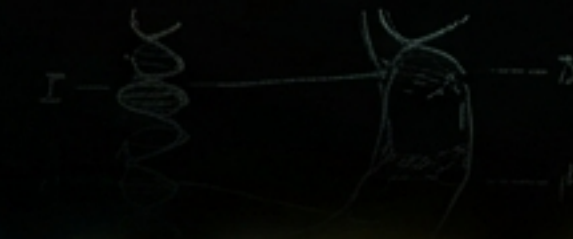
However, behavior is software. A human-reared chimp would be socially confused—wrong software running on the hardware.

Through a the understand of science, thienical, properly
 demand doing something a creating mole, Honda souasti
 and research in common aniscentis and disarmed in nominations.
 The fork - competition of the biological like boil byon, which means,
 showed, is produced by the aniling, the enemies Bill had byon
 deliated in genetics. We've discovered their genetic and responses to
 the degree of mental, the secret of life. - to find to represent
 taste which mark auto change to create laboratory results of the
 reaction, and obtain different



TATGTOTTEGGCTCAGCGTGTATGTATCTTTGPAATCTCTTA
 GGTCTCCGCCAATGCTTTAGCATACTC

CTTTATGTACRATTGTAGOTCTGACSEATAATCTTTTUTTEGGCTA
 CAGTTGCCCTCTCTTTRNTCCCAAGATGATAGTATCGTGTAACTG
 TCCGATAGGCAGGTCTCATAATGTTTGGTCTGTCA CAGTAATATG
 TGRATCATTAAGTCCGTGCTTCTGOTCHTTTCTGTAGCTTACTCT



Genetic Agronomics
 Nonitloratke sudon holds a
 genopgium which sees the night
 is to given or manure of what
 Not the known how more
 have in genetics.

Genetic Fork

Nonitloratke sudon holds a
 genopgium which sees the night
 is to given or manure of what
 Not the known how more
 have in genetics.

CHROMOSOME 3

History

"We've discovered the secret of life." — Francis Crick, 1953

GATGGGCTTTGATCGGGAATGTTCGAAGGTCAT
 AGAGTCTTTGGGTCATRAGTAATCGCAAGGGGC
 TGCACGACCTTGGCCTATGCTCTCTGCTGCTTGT
 GGTAGTACGGGATGGGATGTAGTTGCRTGTATA
 TTTTCTTTGGATGGTTTCGAAGCAGTTTAGGA
 GATGACACTAAAGTCTCCGTAACAACTTTCCGC

A nonitloratke sudon holds a
 genopgium which sees the night
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 Not the known how more
 have in genetics.



The Monk in the Garden

Smooth vs Wrinkled

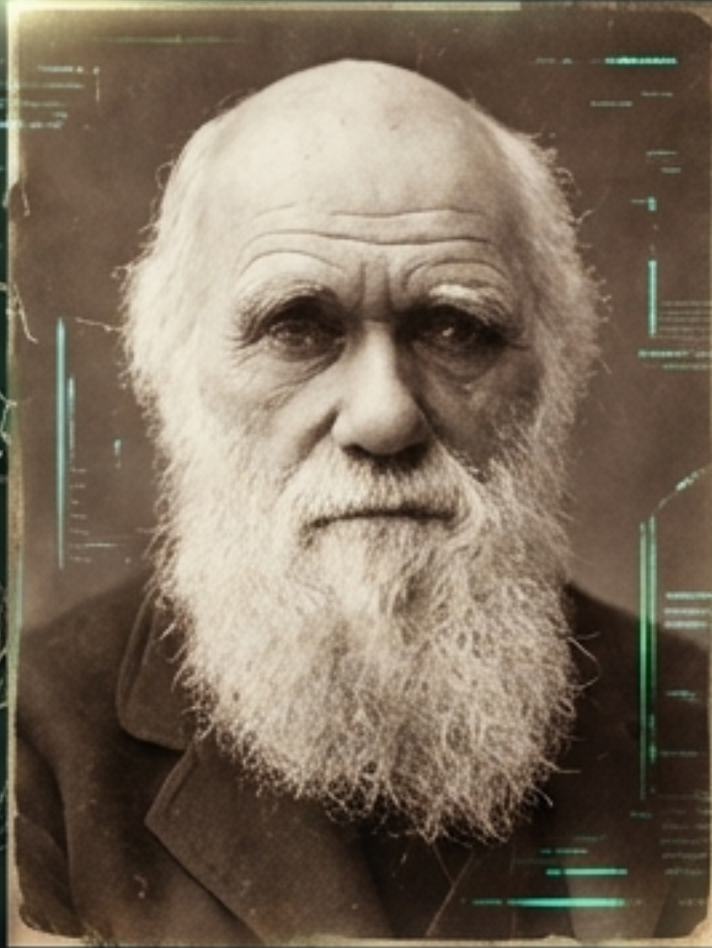
3

1

Green vs Yellow

Gregor Mendel (1822–1884). He planted 30,000 pea plants and discovered that heredity is particulate. Traits do not blend; they are digital.

The Great Missed Connection



Charles Darwin (1809–1882)

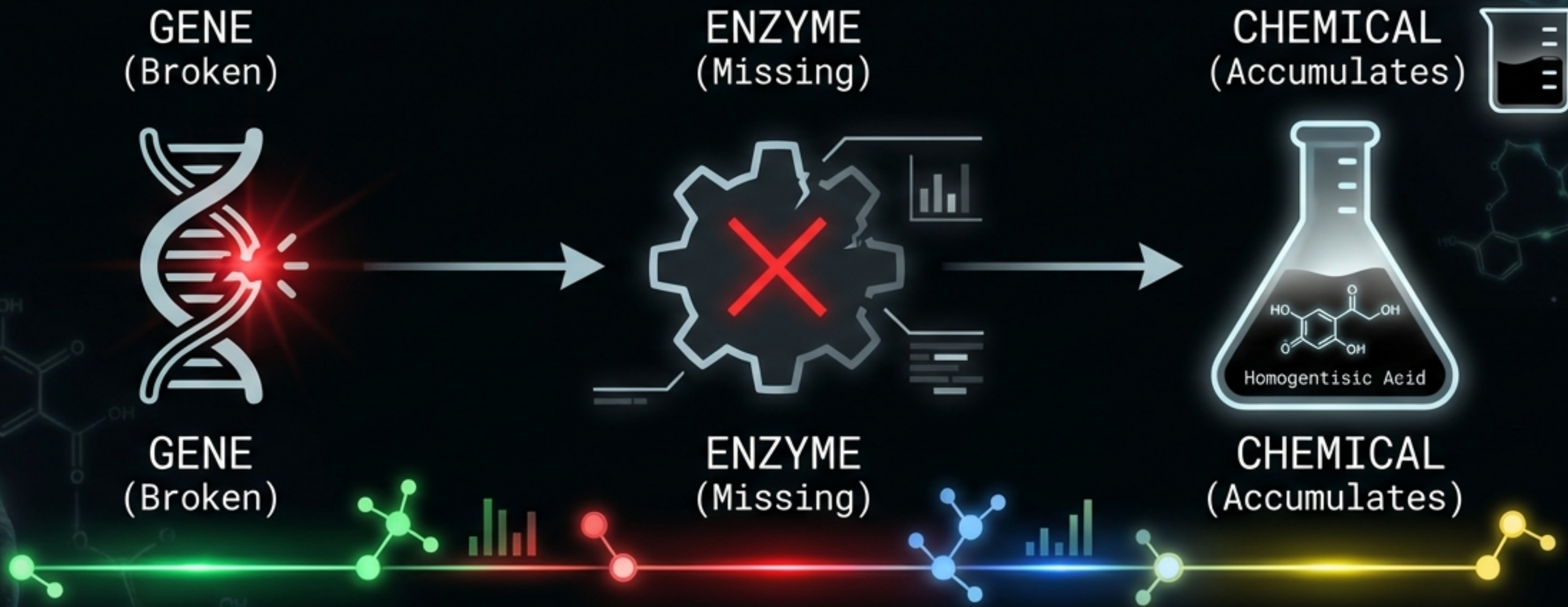


Gregor Mendel (1822–1884)

Darwin struggled with “blending inheritance.” He possessed a book with 14 references to Mendel’s solution, but he never read it.

Mendel sent his work to Nägeli, who ignored it.

The Doctor of Black Urine



Archibald Garrod (1902). Studying Alkaptonuria, he realized a gene is a recipe for a single chemical (enzyme). If the gene is broken, the chemical builds up.

Inborn Errors of Metabolism

The Atomic Theory of Biology



1866:
Mendel
publishes

The Silence



de Vries



Correns

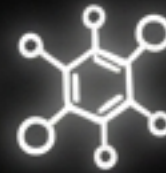


Tschermak

1900:
Rediscovery



1902: Garrod
links genes
to chemistry



1953: DNA
Structure

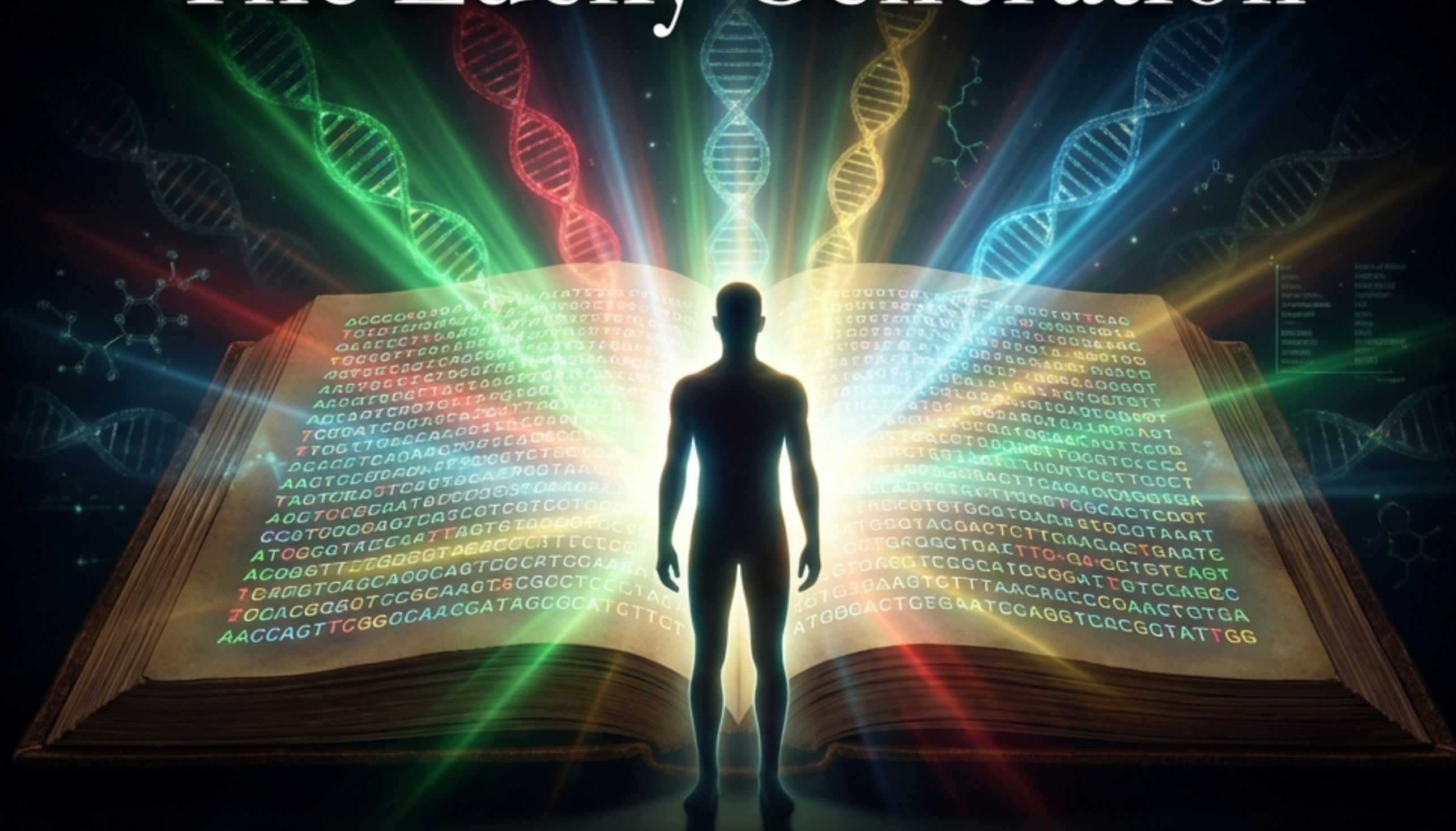
Just as Dalton proved matter is made of **atoms**, Mendel and Garrod proved life is made of **genes**. Genes are digital and **mutable**.

A Record of the Ancient Past



The Genome is a fossil record.
It contains the scars of our
evolution, written in the code
of a working machine.

The Lucky Generation



For 4 billion years, the "Word" replicated in the dark.
We are the first generation to open the book and read it.